Aakash Buddha Kannan

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Summary

Entrepreneurial Computer Science undergraduate focused on turning real-world problems into practical technology products. Experienced in full-stack development, machine learning, and quantum optimization; technical lead for club initiatives. Driven by product thinking, experimentation, and building solutions that create real value.

Education

Rajalakshmi Institute of Technology **Chennai, India**
Bachelor of Engineering in Computer Science *Sep 2024 – May 2028*
- CGPA: 7.8 / 10.0

Achievements

1st Place — Designathon 2025
RIT Digital Twin: Smart Campus Management Platform
- Delivered a smart campus platform featuring JWT authentication, role-based access for 5 user types, timetable generation with PDF export, and classroom booking workflows.
- Stack: React, Spring Boot (Java 21), MySQL 8, Docker; deployed on Vercel and Render with CI/CD configuration.

1st Place — Grand Prix F1 Gaming Tournament Apr. 2026
Gamathon, organized by Revolution Games (Apr. 9–11, 2026)
- Won 1st place in a 3-day competitive F1 gaming tournament.

Participant — Hack The Horizon 2.0, VIT Chennai Aug. 2025
24-Hour Inter-Institutional Hackathon, AI Club, VIT Chennai
- Built and presented a functional prototype in 24 hours at an inter-institutional hackathon.

Projects

FarmSync — Quantum-AI Crop Advisor 2026
Flutter, FastAPI, LightGBM, PennyLane, PostgreSQL

- Built a full-stack crop advisory platform combining agronomic feasibility rules, LightGBM yield prediction, and quantum optimization to recommend crop sequences across multiple growing seasons.
- Formulated crop sequencing as a QUBO and implemented a 12-qubit SPARQ optimization pipeline using PennyLane with constraint-preserving XY mixing, warm-start initialization, and measurement-based ranking.

Sevagan — Civic Issue Tracking Platform 2026
React, Express, Supabase, PostgreSQL, Tailwind CSS, Solidity

- Developed a role-based civic issue platform with citizen, admin, and worker workflows for issue reporting, assignment, status tracking, and completion-photo management.
- Built REST APIs with Express and Supabase/PostgreSQL persistence, including image uploads and an optional Ethereum/Hardhat layer for issue hashing and integrity verification.

Infinitus — Mathematics Club Platform 2026 – Present
Next.js, React, Supabase, PostgreSQL, Tailwind CSS, Resend

- Developing the official mathematics club platform with reusable Next.js components, responsive event/member pages, contact workflows, and an authenticated admin dashboard.
- Implemented server-side API routes with Supabase/PostgreSQL persistence, event registration-link management, contact-message handling, rate limiting, and email notifications through Resend.

Irrumbu Kottaai Pallupudingi — 2D Action Platformer 2026 – Present
Godot 4, GDScript

- Engineered responsive movement, combat, enemy finite-state machines, and predictive multi-phase boss AI, with phase transitions, combo attacks, lunge behavior, and centralized damage handling.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C/C++, SQL
Frontend: React, Next.js, Vite, Tailwind CSS, React Router
Backend: FastAPI, Spring Boot, Node.js, Express, REST APIs
Databases: PostgreSQL, MySQL, SQLite, Firebase/Firestore, Supabase
Tools: Git, Docker, Maven, Vercel, Render, Hardhat, Godot 4

Extra-Curricular

Technical Head — Infinitus Club, RIT: Leads technical development and digital initiatives, including web platforms and technical decisions.

Editor — Women's Empowerment Club, RIT: Created content promoting awareness and student engagement.

Volleyball (Spiker): Represented School in competitive matches; developed discipline and team coordination.